Palindrome.

Brute Force :- TC: O(n + n + n) — create a copy, reverse the copy, Compare.

AS: O(n).

H1
1 → 2 → 3 → 2 → 1 → N.
t1 t1 t1 t1 t1 t1

H2
1 → 2 → 3 → 2 → 1 → N.
t2 t2 t2 t2 t2 t2

1 → 2 → 3 ⁝ → 3 → 2 → 1 → N.

1 → 2 → 3 → 2 → 1 → N.

A palindrome mirrors around its centre.

H.
1 → 2 → 3 ⁝ → 3 → 2 → 1 → N.

H1
1 → 2 → 3 → N.

H2
3 → 2 → 1 → N.

reverse.

H2
1 → 2 → 3 → N.

TC: O(n).
AS: O(1).

H.
i/p: 1 → 2 → 3a → 3b → 2 → 1 → N.

o/p: H.
1 → 2 → 3 → N.

H2
3 → 2 → 1 → N.

H. t. H2
1 → 2 → 3a → 3b → 2 → 1 → N
N.

H2 = t → next
t → next = N.

s s s
1 → 2 → 3 → 3 → 2 → 1 → N.
f f f

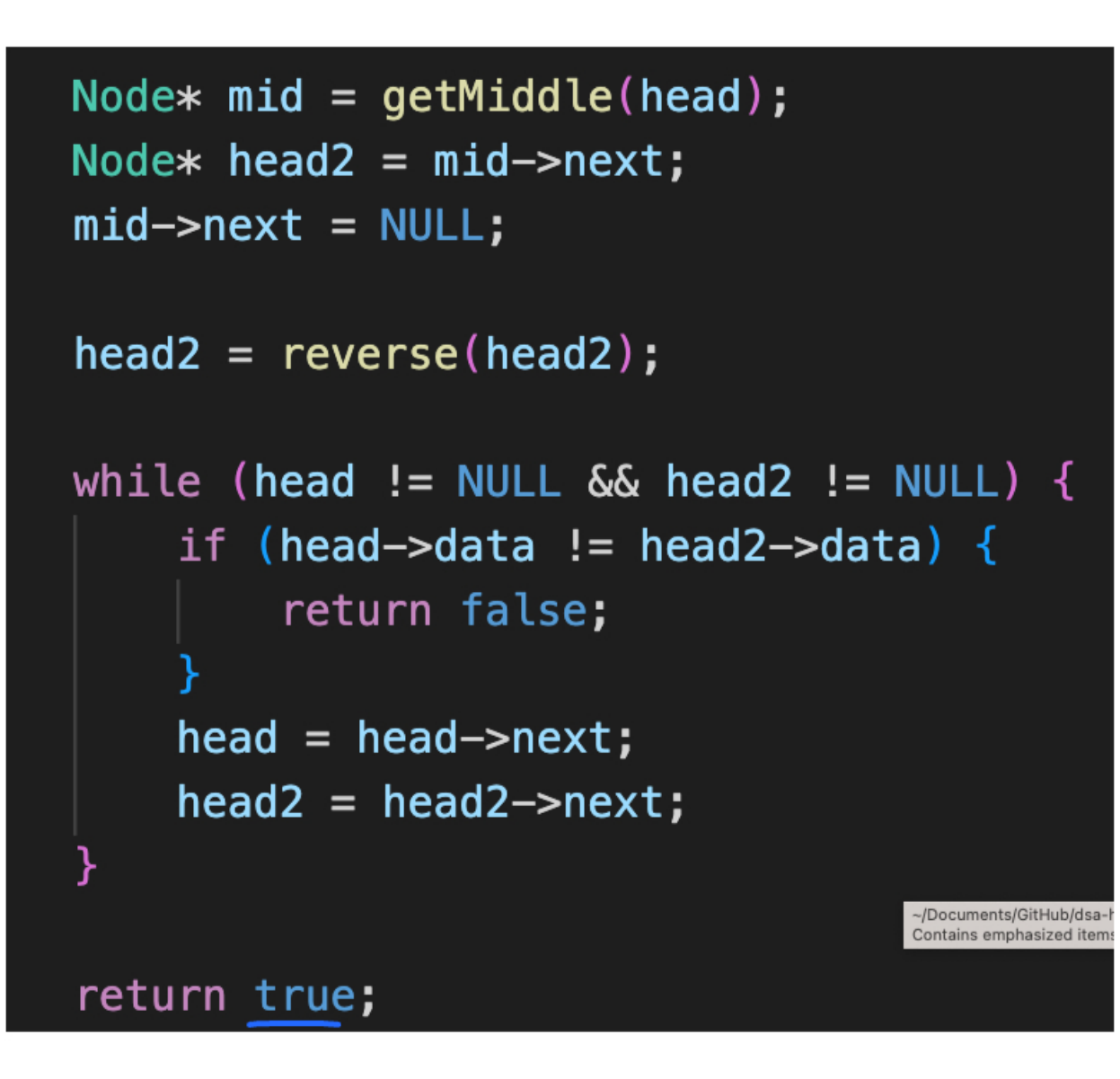
f → next → next == NULL

H. mid.
1 → 2 → 3 → 2 → 1 → N.

H.
1 → 2 → 3 → N.

H2
2 → 1 → N.

H2
1 → 2 → N.



```
Node* mid = getMiddle(head);
Node* head2 = mid->next;
mid->next = NULL;

head2 = reverse(head2);

while (head != NULL && head2 != NULL) {
    if (head->data != head2->data) {
        return false;
    }
    head = head->next;
    head2 = head2->next;
}

return true;
```

~/Documents/GitHub/dsa-
Contains emphasized item

H. H2
1 → N.
mid.

Merge 2 sorted LL.

H1
5 → 10 → 15 → 40 → N.

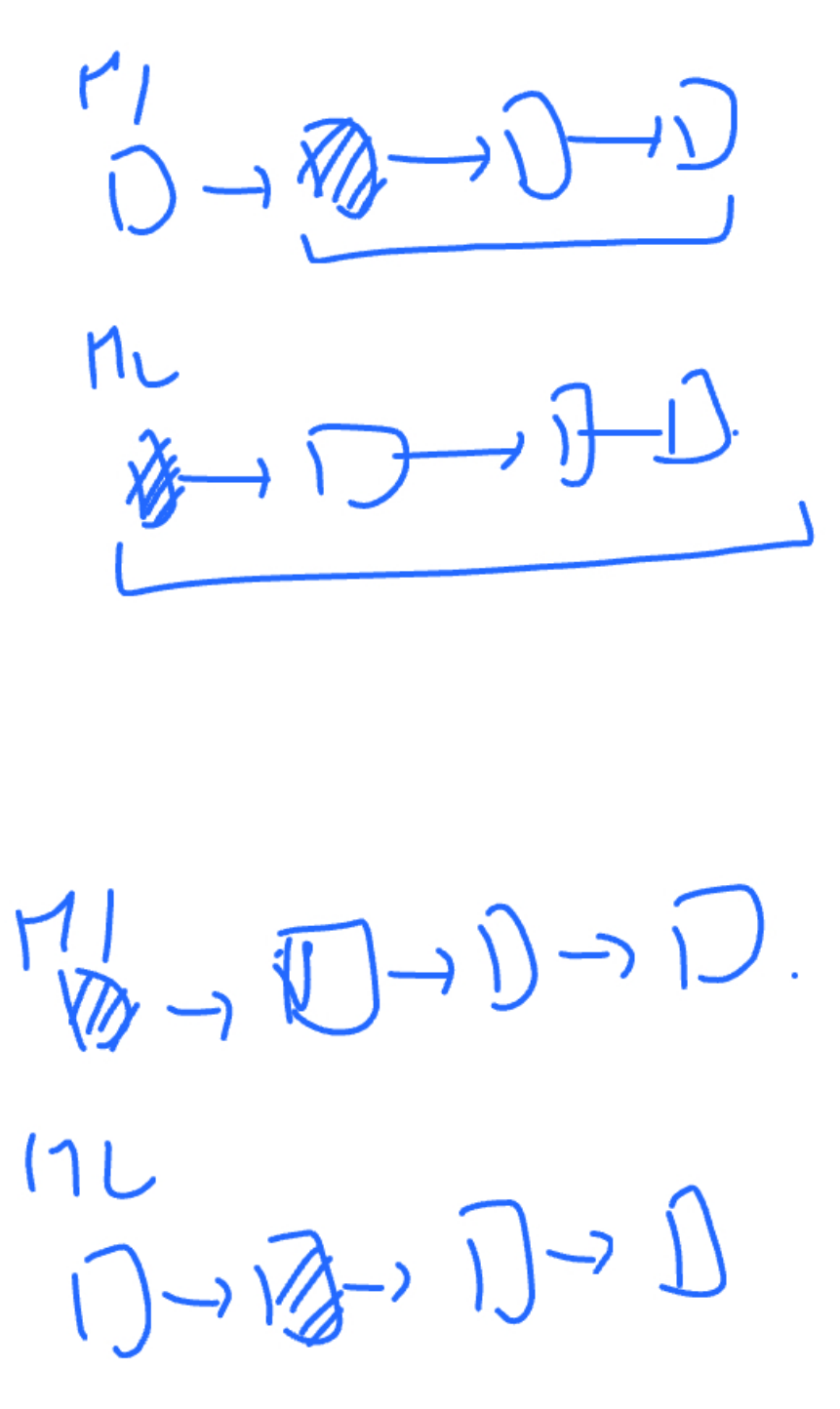
H2
2 → 3 → 20 → N.

o/p: H.
2 → 3 → 5 → 10 → 15 → 20 → 40 → N.

H1
5 → 10 → 15 → 40 → N.

H2.
3 → 20 → N.

t
2

H.
3 → 5 → 10 → 15 → 20 → 40 → N